VARUN CHITTMELLA

Cincinnati, Ohio | chittmvn@mail.uc.edu | 513-399-0196 | LinkedIn | https://github.com/Varrray

EDUCATION

Masters in Computer Science **May 2023-Apr 2025**

University of Cincinnati, Cincinnati, Ohio

Coursework: Advance Algorithms, Database Theory, Machine Learning, Advanced Database Management , Cloud Computing, Deep Learning

Bachelor of Technology in Computer Science Engineering **Jun 2017-Aug 2021**

GITAM University, Hyderabad, India

SKILLS

- **Backend & Languages:** Java 17, Spring Boot, Spring Cloud, REST APIs, GraphQL, SQL, Python, JavaScript, C/C++
- **AI/ML & Data:** Machine Learning, Deep Learning, RAG, LangChain, LangGraph, Vector Stores
- **Cloud & DevOps:** Docker, Kubernetes, AWS (EC2, S3, RDS), Git/GitHub, GitHub Actions, CI/CD
- **Data & Messaging:** PostgreSQL, MongoDB, Redis, Kafka, Elasticsearch
- Architecture & Performance: Microservices, Distributed Systems, Event-Driven Architecture, Low-Latency Systems
- **AI & LLM Integration:** OpenAI Chat Completions API, prompt engineering, retrieval-augmented generation (RAG)

WORK EXPERIENCE

Software Engineer, York Solutions LLC,Cincinnati,USA **Apr 2025 - Present**

- Designed a SaaS AI coaching platform delivering stochastic, personalized learning journeys by tuning content flow and difficulty, resulting in higher user engagement and retention.
- Built a full-duplex voice assistant pipeline with real-time VAD, streaming ASR, and TTS, achieving <500 ms end-to-end latency for natural, uninterrupted conversations.
- Formulated event-driven microservices with Kafka and Redis, streamlining orchestration and boosting message throughput by 30%.
- Refactored binary data pipelines to stream bytea payloads in chunks and enhance metadata indexing, speeding up upload/download operations by 30%.
- Utilized LangChain, LangGraph, and MCP to coordinate AI reasoning workflows, cutting manual prompt-tuning effort by 40% and improving relevance scores by 25%.
- Expanded the AI service orchestration tool, raising coaching accuracy by 18%, while integrating GraphQL services to manage journeys, dimensions, and XP tracking—advancing data-driven insights by 22%.
- Rolled out a real-time feedback analytics pipeline analyzing 5K+ monthly ratings, enabling iterative prompt enhancements that amplified user satisfaction by 17%.

Software Engineer Intern, Resilient Communities, Cincinnati , USA **Oct 2024-Apr 2025**

- Constructed an intelligent real-estate evaluation engine that unified county-level datasets with multi-platform property feeds (Zillow, Redfin, Realtor.com, Franklin County Auditor), enabling highly accurate asset discovery and valuation.
- Choreographed multi-agent LangChain/LangGraph architectures that streamlined cross-source ingestion, harmonized metadata, and generated investment metrics (GRM, NOI, Cap Rate) with 92% precision, accelerating decision-making by 50%.
- Applied Model Context Protocol (MCP) to harmonize LLM interactions across distributed components, facilitating interchangeable AI modules for acquisition analysis, renovation cost forecasting, and rent modeling.
- Introduced photo-based rehabilitation assessment via AI-driven image interpretation, delivering 85% accurate repair estimates and reducing manual evaluation cycles by 35%.
- Implemented a React/Next.js analytic console presenting zoning overlays, geographic insights, and financial dashboards, elevating investor intelligence workflows by 40%.
- Generated PDF/CSV investment briefs and interactive pin-based property mapping, shrinking reporting overhead by 70% and improving stakeholder visibility.
- Refined end-to-end processing chains with optimized transformations, slashing manual analysis demand by 60% and strengthening model-backed recommendation reliability.

Software Engineer, Tata Consultancy Services, Hyderabad, India **Oct 2021 – July 2023**

- Architected a large-scale Customer Management & Orchestration ecosystem supporting 5M+ subscribers for a major US cable provider, streamlining lifecycle operations including new activations, renewals, and service disconnects. Engineered Spring Boot microservices for order execution, workflow automation, and customer status tracking, integrating REST APIs with Kafka-driven asynchronous processing.
- Enhanced platform performance by embedding Redis-based caching, refining SQL queries, and restructuring high-volume endpoints—achieving a 25% reduction in average latency. Implemented resilient execution patterns such as circuit breakers, retry logic, and idempotency safeguards, strengthening fault tolerance across distributed transactional flows.
- Partnered with frontend, QA, and product teams to deliver 15+ high-stability releases through CI/CD pipelines, robust testing practices, and strict code-quality enforcement (SonarQube, Checkstyle). Advanced two major system modernization initiatives that elevated overall reliability by 28% while cutting operational workload by 18%.
- Designed and executed 50+ targeted test cases, boosting code efficiency by 25% and validating performance across 10+ critical workloads.
  ensuring robust performance across 10+ scenarios.
- Achieved a 40% decrease in production defects through systematic quality improvements and validation strategies.
- Delivered automation capabilities that minimized recurring system exceptions by 30%, strengthening platform consistency and maintainability.

Software Engineer Trainee, Tata Consultancy Services, Hyderabad, India **Jun 2021 – Oct 2021**

- Supported backend development for customer onboarding features using Java, Spring Boot, and REST APIs.
- Integrated backend APIs with React-based UI, improving data-fetching efficiency and reducing API response time by 20%.
- Rolled modular components and participated in Agile ceremonies, contributing to faster delivery cycles.

Executed unit and integration tests to ensure functionality and reliability of new backend services.

Facial Landmark Detection

- Achieve a facial feature categorization system that lifts user experience by ensuring accurate representation and analysis of facial attributes
- Used RNN to create the model and used Resnet-50, which has an accuracy of 85%.
- Technologies: Python (SKLearn, NumPy, Pandas), HTML, CSS, JavaScript

CERTIFICATIONS AND COURSES

- **Oracle certified Professional Java SE 17 Developer**